Evaluation of a reduced Schlögl reaction pathway signal rectifier in kinetic simulations

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Abstract

A deterministic simulation is used to evaluate the bistable behaviour of a reduced signal rectifier by demonstrating amplification and dilution above and below a critical threshold. A stochastic simulation with the same rectifier is implemented to demonstrate the influence that noise produced by the simulation algorithm has on small system sizes, by calculating the probability that a simulation correctly rectifies the concentration at given initial thresholds at varying system sizes. A set of DSD gates is also proposed that theoretically equate to the proposed further reduced pathway.

Definitions

Mass Action

Mass action dictates that the rate of a reaction is proportional to the concentration of reactants in solution.[2]

Kinetic Simulations

Kinetic simulations calculate the results of a chemical reaction (where molecules collide with one another and react). Given a specified set of chemical species, their initial concentrations, reactions between species, and the rates of the reactions occurring between the specified species, concentrations change throughout a simulation in response to the given rates and initial concentrations.

Signal Rectifiers

Signal rectifiers provide the logical behaviour of a given pathway (the actual reactions themselves), and these are used in tandem with kinetic simulations to simulate reaction pathways over a specified duration.

Reaction States

Reaction pathways have specific states that influence their behaviour. Monostable reaction pathways only have one stable state, which produces the same outcome each time. Bistable reaction pathways have two stable states (x_{low} and x_{high}) and one unstable state that is between the two (x_{crit}), which is unstable as initial concentrations above and below x_{crit} influence which stable state the reaction leads to.

Schlögl Reaction

Original Pathway

$$2\,\mathrm{X} + \mathrm{A} \xrightarrow{k_1} 3\,\mathrm{X} \tag{1}$$

$$3\,\mathrm{X} \xrightarrow{k_2} 2\,\mathrm{X} + \mathrm{A} \tag{2}$$

$$\mathrm{B} \xrightarrow{k_3} \mathrm{X} \tag{3}$$

$$\mathrm{X} \xrightarrow{k_4} \mathrm{B} \tag{4}$$

The Schlögl Reaction is a bistable reaction pathway that uses two fuel types (A and B) that are assumed to be in constant supply by a reservoir, along with the dynamic species X, which has the stoichiometric counts 2X and 3X. Through its own autocatalytic reduction and production, two stable states are produced in which the concentration of X is amplified to (x_{high}) or diluted to (x_{low}), and an unstable state where the concentration of X does not change (x_{crit}). There is a reaction that converts X to a product B, giving a sink that reduces the concentration of X.[5]

Reduced Pathway

This evaluation uses a reduced Schlögl pathway where the reactions involving the dynamic species X still occur, but k3 is assumed to be 1 to create a constant sink, and S is assumed to be in constant supply.[7]

$$2\,\mathrm{X} + \mathrm{S} \xrightarrow{k_1} 3\,\mathrm{X} \tag{5}$$

An X is added to the concentration by 2X reacting with S to form 3X.

$$3\,\mathrm{X} \xrightarrow{k_2} 2\,\mathrm{X} + \mathrm{S} \tag{6}$$

An X is removed from the concentration when 2X and S are produced by reducing 3X.

$$\mathrm{X} \xrightarrow{k_3} \mathrm{W} \tag{7}$$

An X is removed from the concentration by producing W.

$$2\,\mathrm{X} \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 3\,\mathrm{X} \tag{8}$$

$$\mathrm{X} \xrightarrow{k_3=1} \mathrm{W} \tag{9}$$

The reaction is further simplified.

Deterministic Simulations

Deterministic kinetic simulations calculate concentration changes within a reaction using an Ordinary Differential Equation (ODE), providing mathematically determined results of a reaction over a specified time period.

$$\frac{dx}{dt} = k_1 A x^2 - k_2 x^3 + k_3 B - k_4 x \tag{10}$$

This is the ODE of the original, unreduced Schlögl reaction pathway.[5][7]

$$\frac{dx}{dt} = k_1 x^2 - k_2 x^3 - x \tag{11}$$

Since a reduced pathway is used, the ODE is also reduced.

Stochastic Simulations

Stochastic simulations use propensities rather than reaction rate constants; these propensities represent the chance that a given reaction will occur within the next calculation. These pseudo-random reaction events provide a more realistic representation of reaction kinetics, where the exact perfect conditions don't always occur, and randomly occurring reactions can significantly change the result of a chemical reaction.[1][2]

DNA Strand Displacement

DNA Strand Displacement occurs when the toehold (binding site) of a single invading DNA strand replaces another strand by initially binding to the toehold of a partially double-stranded DNA complex to initiate branch migration, which is the process of the invading strand exchanging base pairs with the incumbent strand, until it replaces the incumbent strand fully.[4][6]

Methodology

Threshold Determination

The reduced ODE can be manipulated using arithmetic, through factorization and reconstitution as a quadratic, and then substituted into the quadratic formula.

$$\frac{dx}{dt} = k_1 x^2 - k_2 x^3 - x \tag{12}$$

This is the reduced ODE introduced earlier.

$$\frac{dx}{dt} = 0 \tag{13}$$

The reaction is at equilibrium when dx/dt is equal to zero.

$$0 = k_1 x^2 - k_2 x^3 - x \tag{14}$$

$$0 = x(k_1 x - k_2 x^2 - 1) \tag{15}$$

x can be factored out. Therefore x_{low} (the stable state) is always 0.

$$k_1 x - k_2 x^2 - 1 = 0 \tag{16}$$

And then further rearranged.

$$-k_1 x + k_2 x^2 + 1 = 0 \tag{17}$$

The rearranged equation is then multiplied by -1.

$$k_2 x^2 - k_1 x + 1 = 0 \tag{18}$$

Yielding the form of a simple quadratic.

$$x_{\text{crit}} = \frac{k_1 - \sqrt{k_1^2 - 4\,k_2}}{2\,k_2} \tag{19}$$

$$x_{\text{high}} = \frac{k_1 + \sqrt{k_1^2 - 4\,k_2}}{2\,k_2} \tag{20}$$

This simple quadratic can then be substituted into the quadratic formula, providing two additional calculable values that are the x_{crit} and x_{high} thresholds for a given pair of k_1 and k_2 reaction rates.[7]

Evaluation

Deterministic Simulation

A signal rectifier and functions that equate to the reduced differential equation and the calculations previously introduced to determine the critical and high threshold were implemented as functions in Python. The signal rectifier is provided as a parameter to the deterministic simulation algorithm supplied by the SciPy module, which simulates the kinetics of the ODE function.

Amplification and Dilution

The constructed rectifier's output was initially experimented with using a critical threshold calculated with the reduced ODE to establish if a critical concentration threshold is produced.

$$x_{\text{crit}} = \frac{2 - \sqrt{2^2 - 1.6}}{0.8} = 0.563508 \tag{21}$$

The calculated x_{crit} threshold for the reaction rates k_1=2 and k_2=0.4.

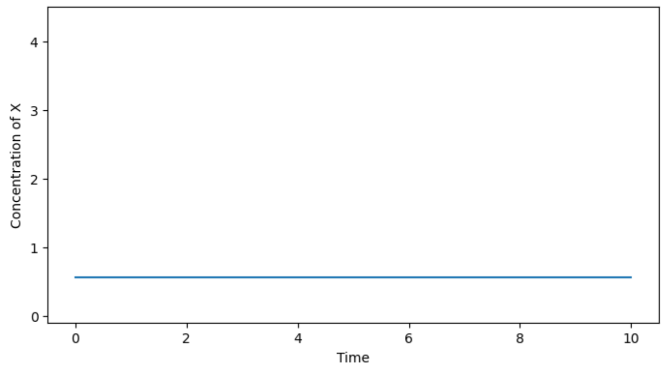

Figure 1: An unchanging X concentration at (x_{crit})

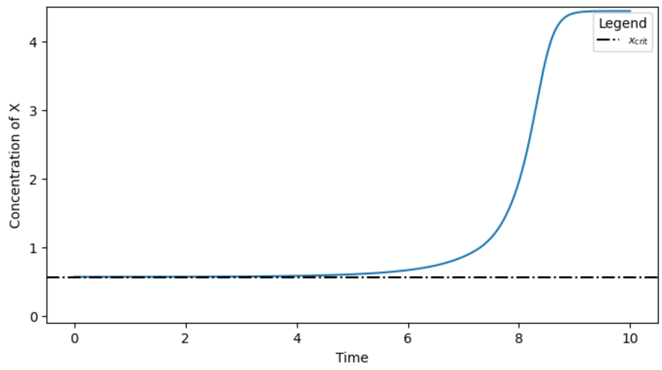

Figure 2: X Concentration amplification with an initial concentration above x_{crit} (0.564)

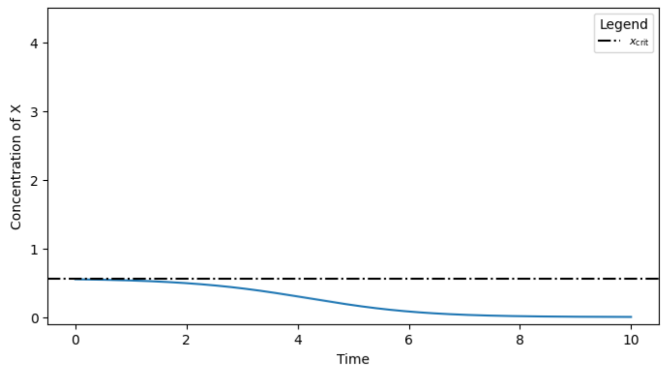

Figure 3: X Concentration dilution with an initial concentration below x_{crit}. (0.55)

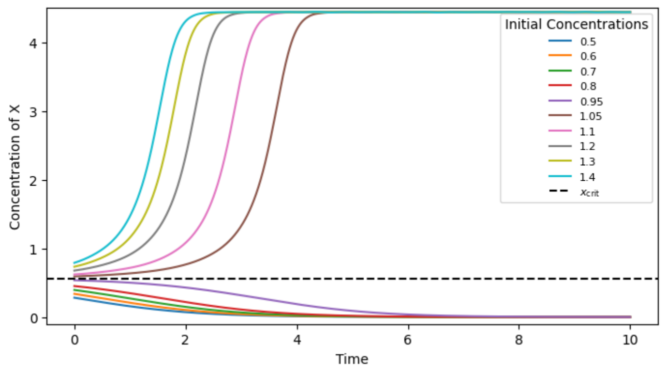

Figure 4: Bistable activity of various initial concentrations.

Each simulation used the reaction rate k_1=2 and k_2=0.4. Figure 1 demonstrates the unstable critical threshold (x_{crit}), where the reaction is at constant equilibrium. Figures 2 and 3 demonstrate that initial concentrations lower and higher than x_{crit} result in the amplification of the concentration of X when above the critical threshold ($x(0) > x_{crit}$) and dilution of the concentration of X when below the critical threshold ($x(0) < x_{crit}$). By iterating through a list of varying initial thresholds, determined results were simulated and plotted on a graph, producing Figure 4, which further demonstrates the behaviour shown in Figures 2 and 3 with additional initial concentrations.

Rate Dependence of x_{crit}

Reaction Speed

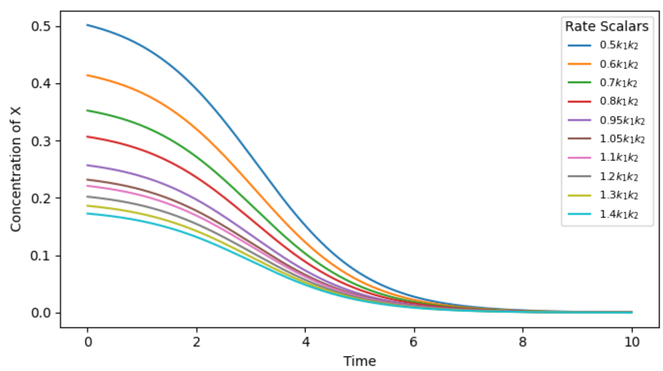

Figure 5: Various k1 and k2 reaction rates scaled with the initial concentration of $0.95x_{crit(k_1,k_2)}$ (scaled from k_1=2, k_2=0.4)

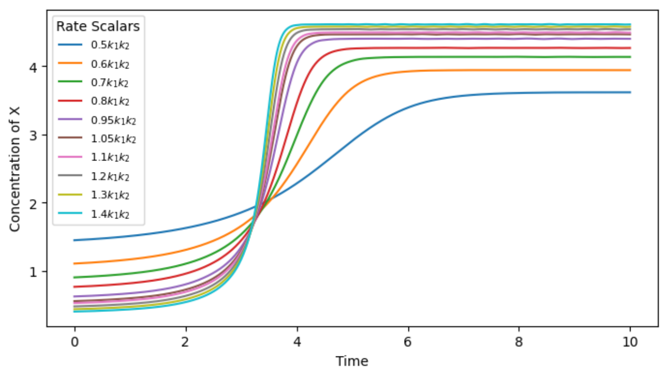

Figure 6: Various k1 and k2 reaction rates scaled with the initial concentration of $1.05x_{crit(k_1,k_2)}$ (scaled from k_1=2, k_2=0.4)

When given larger and smaller reaction rates, the rate at which the concentration climbs is either faster or slower, but the same critical threshold behaviour still occurs, as shown in Figures 5 and 6.

Monostability

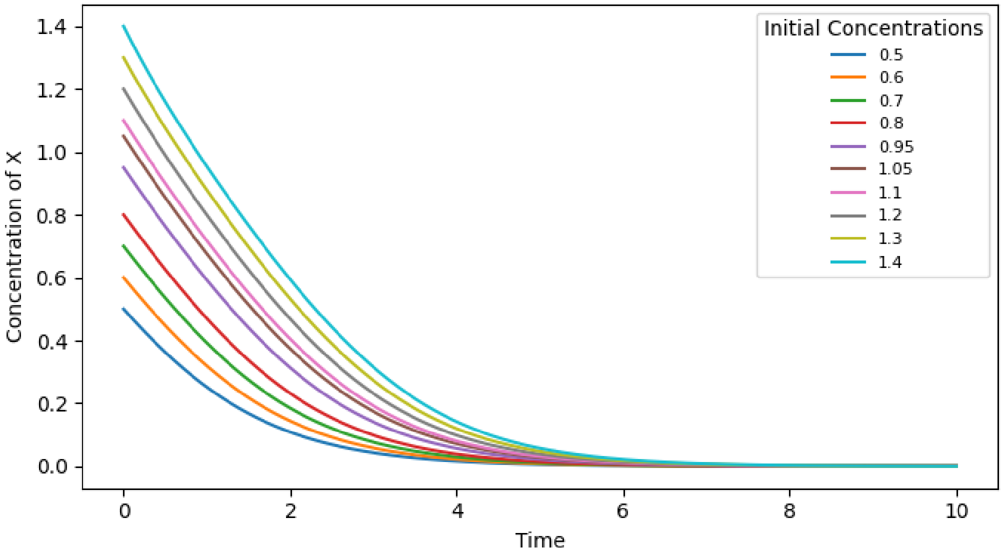

Figure 7: Monostable activity of various initial concentrations, plotting the concentration change of X over time, caused by reaction rates that do not allow for bistability.

Some reaction rates result in the output becoming monostable, as shown in Figure 7, where simulations result in the concentration of X always diluting when the rectifier is provided the reaction rates k_1=1 and k_2=0.4.

A mathematical explanation of this behaviour is that the quadratic shown earlier doesn't produce real roots when calculating the discriminant of the quadratic formula with some reaction rates, resulting in the lack of derivable x_{crit} and x_{high} thresholds, leaving only the state x_{low}. Thus, reaction rates must be $k_1^2 - 4k_2 > 0$, to be bistable.[7]

Stochastic Simulation

Propensity Adjustment

The propensity of reactions in a stochastic simulation must be correctly scaled when changing system size; this is done by accounting for the total permutations of dynamic species combinations that a reaction uses proportionately to a given system size. A symmetry factor that accounts for this can be derived from combinatorics and is used later to correctly scale the propensity for differing system sizes and reaction rates.[1][2]

$$\binom{n}{2} = \frac{n(n-1)}{2} \tag{22}$$

For reactions that use 2 of X, the symmetry factor is 2.

$$\binom{n}{3} = \frac{n(n-1)(n-2)}{6} \tag{23}$$

For reactions that use 3 of X, the symmetry factor is 6.

Amplification and Dilution

The provided Stochastic Simulation Algorithm requires a set containing two tuples that represent reactions and products, and the propensity rate of that reaction, which is the signal rectifier. Adding an extra species to the reactant or product tuple increases the amount of that species required or produced in a reaction.

To simplify testing, a function was written that creates a set of simulation parameters by combining a set of reactions that follow the pathway with their correctly scaled propensities. Comparative thresholds were pre-determined using the previously implemented function for calculating a critical threshold with the ODE.[1]

To evaluate that initial concentrations marginally above and below the critical threshold ($1.05x_{crit}$, $0.95x_{crit}$) were correctly amplified and diluted when compared to the determined result, simulations were run with varying system sizes and their final result compared with the determined result to check if the concentration was correctly amplified or diluted, which was calculated by comparing if the result is lower or higher than a calculated midpoint ($x_{high}/2$). Each system size was simulated 500 times to gather accurate probabilities. Each simulation used the reaction rates k_1=2 and k_2=0.4.

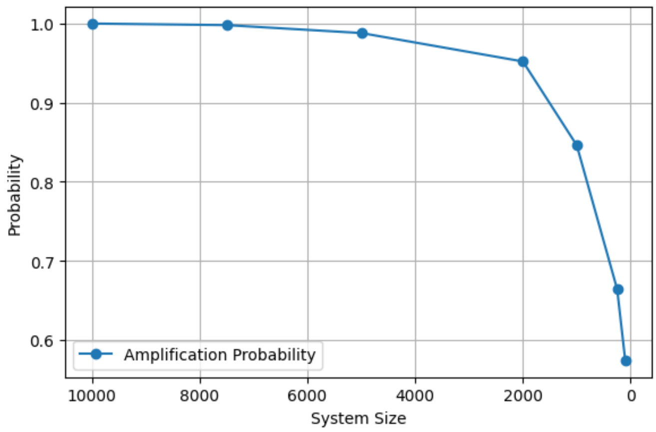

Figure 8: Probability of the initial concentration to be amplified compared by system size

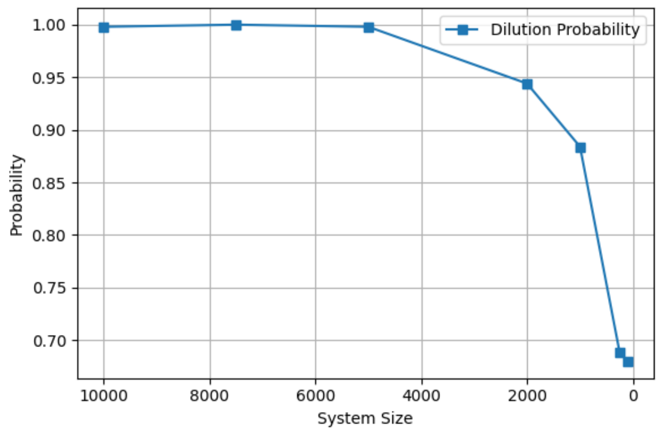

Figure 9: Probability of the initial concentration to be diluted compared by system size

Given the initial concentration $1.05x_{crit}$, the simulation yielded a probability of 1 to correctly amplify the concentration of X at system size 10,000, but only yielded the probability of 0.68 at system size 100. (Figure 8)

Given the initial concentration $0.95x_{crit}$, the simulation yielded a probability of 1 to correctly dilute the concentration of X at system size 10,000, but only yielded the probability of 0.574 at system size 100. (Figure 9)

Stochastic simulations are prone to noise due to the inherent randomness of the stochastic simulation algorithm. This prevents the determined behaviour from occurring in small system sizes where random shifts in concentration can cause distortion that skews the predicted behaviour, which can be observed in Figures 8 and 9, where the probability of correctly amplifying and diluting the concentration of X is reduced as the system size decreases.

DNA Displacement Simulation

For the proposal and demonstration of DSD gates, a further reduced pathway is used.

$$2\,\mathrm{X} \xrightarrow{k_1} 3\,\mathrm{X} \tag{24}$$

$$\mathrm{X} \xrightarrow{k_3=1} \mathrm{W} \tag{25}$$

Proposed Gates

The DSD tool has built-in example gates which can partially equate to the reactions in the further reduced Schlögl pathway.[3] In the context of a DNA displacement simulation, X would be an identical DNA strand and W a different strand.

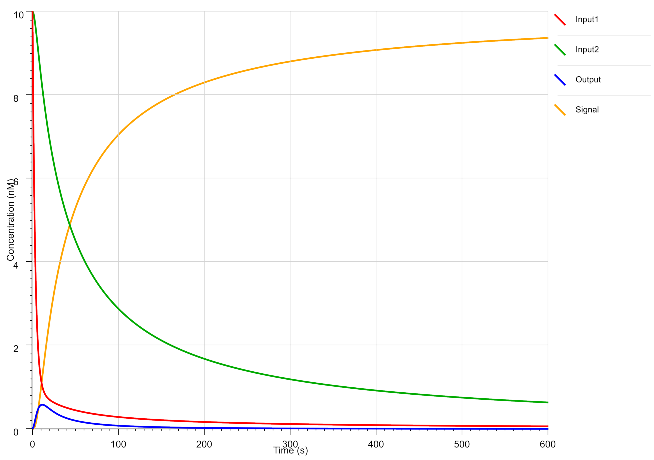

Figure 10: Concentration changes of the Join gate.

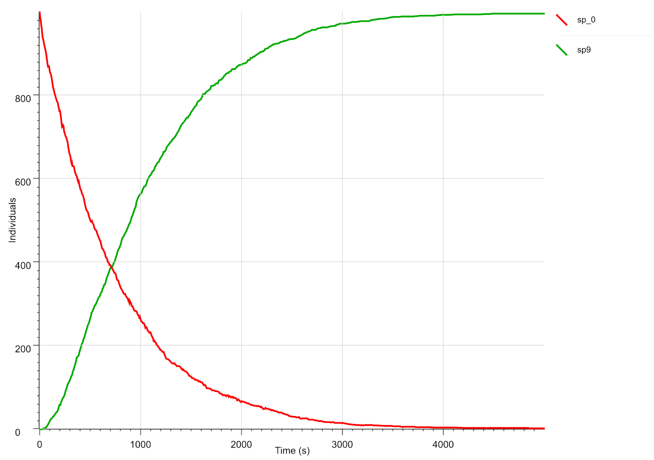

Figure 11: Concentration changes of the Transducer gate.

A join gate could be used to partially reproduce the autocatalysis of $2\,\mathrm{X} \xrightarrow{k_1} 3\,\mathrm{X}$; this gate requires two strands to release the third strand. Initially, this gate exposes one toehold, which, upon binding, initiates branch migration. After branch migration, the second toehold is exposed for an additional strand to bind to, which releases the strands alongside an additional strand. Introducing a fuel species wouldn't be effective due to the lack of a constant supply of fuel to drive autocatalysis, because of this DSD simulations cannot be truly bistable.

A transducer gate could be used to reproduce the sink produced by X $\xrightarrow{k_3=1}$ W, this gate binds an X to create W by initially binding X to an exposed toehold, then performing branch migration to replace the incumbent strand and release W, X is held within the gate removing it from concentration.

Rate constant adjustment

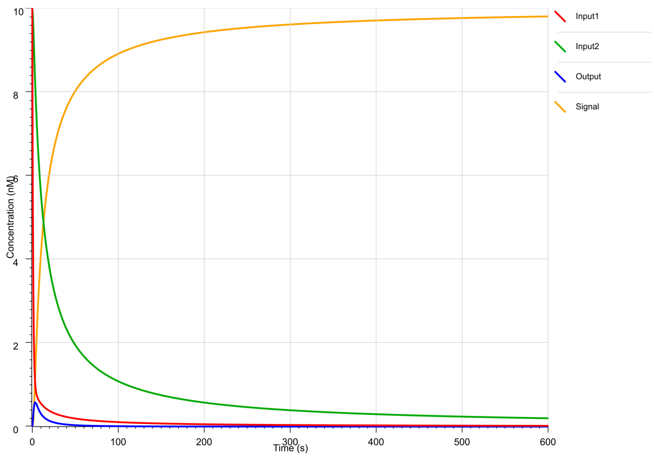

Figure 12: Concentration changes of the Join gate when using the directive parameters [k=0.01, u=1]

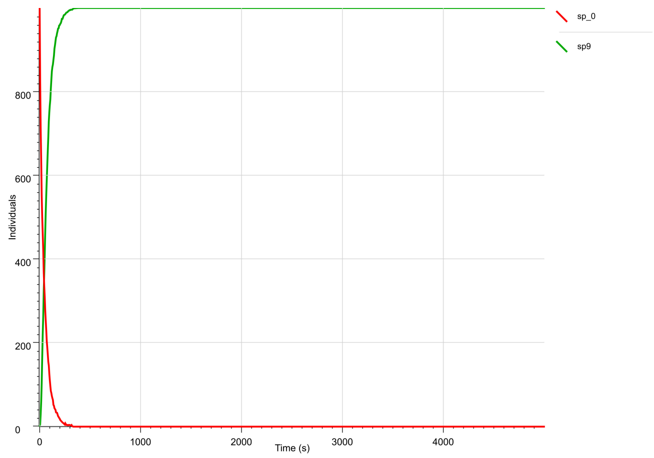

Figure 13: Concentration changes of the Transducer gate when using bind=0.005 and unbind=0.1126

Differing rate constants can be achieved in the proposed gates through adjusting the toehold binding rate in the directive parameters or the defined rate variables, as shown in Figures 12 and 13. [8]

Conclusion

In summary, the constructed signal rectifiers based on the reduced Schlögl reaction pathway do show bistable behaviour (Figure 4), but only when given compatible reaction rates (Figure 7). The concentration of X is correctly amplified and diluted with initial concentrations above and below the critical threshold, following mass action (Figures 1 2 3). The rate at which the concentration shift occurs is relative to the given reaction rates (Figures 5 6). Stochastic simulations behave randomly and do not always provide the determined result due to the noise produced, and therefore, they must be simulated multiple times when using smaller system sizes to derive an accurate result (Figures 8 9). Although the further reduced pathway is not demonstrated, two gates are proposed that could theoretically be used in tandem to partially replicate the reduced reaction pathway (without true bistability) in a DNA displacement simulation (Figures 10 11); the reaction rate constants can also be adjusted through changing the toehold binding rates (Figures 12 13).

References

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